

Evaluating Hybrid Sparse-Dense Retrieval for EU Climate Legal Documents

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Abstract—Legal information retrieval in statutory corpora poses specific challenges due to formulaic language, hierarchical authority structures, and the need for doctrinal precision. Existing hybrid legal-retrieval studies focus on case law or large general corpora, and none controls for vocabulary leakage between evaluation queries and the indexed statutes, which we show can single-handedly determine which retrieval paradigm appears superior. This paper evaluates three first-stage retrieval configurations on a 102-query benchmark over EU Climate Law: BM25 lexical search, `nomic-embed-text-v1.5` dense retrieval (768-dim, 8,192-token context), and a hybrid combining both via weighted Reciprocal Rank Fusion (RRF, $k=20$, dense weight 5). Of these queries, 31 in the test set are derived from real written questions submitted by Members of the European Parliament, providing externally sourced information needs that control against query-corpus vocabulary leakage. The corpus comprises 1,156 article-level provisions (1,166 retrieval units after splitting oversized amendment articles) extracted from 72 EU legislative instruments via SPARQL and Formex XML. Fusion parameters were tuned on a 49-query validation split; all reported metrics are from the sealed 53-query test set. Dense retrieval achieves $HR@5=0.925$, substantially outperforming BM25 ($HR@5=0.698$; Wilcoxon $p=0.004$). The hybrid achieves $HR@5=0.906$, $MRR@5=0.773$, and $NDCG@5=0.721$, the highest MRR and NDCG of all three systems, but its improvement over dense alone is statistically non-significant ($p=0.614$). Only 5 of 53 queries (9.4%) are hard failures for the hybrid. A paired experiment re-evaluating 71 identical information needs under their original corpus-derived wording shows why leakage control matters: reworded queries cost BM25 0.29 of $MRR@5$ but the dense retriever only 0.09, meaning leaky benchmarks manufacture a false parity between lexical and semantic retrieval. The results confirm that a strong general-domain dense retriever already approaches ceiling performance on this corpus, and that weighted lexical-semantic fusion provides modest ranking gains without statistically significant coverage improvement.

Index Terms—Legal Information Retrieval, Hybrid Retrieval, BM25, Dense Retrieval, Reciprocal Rank Fusion, EU Climate Law, Retrieval-Augmented Generation, Statute-Level Benchmark

I. INTRODUCTION

Anyone who has tried to answer a legal question from primary legislation knows that finding the right document is most of the battle. The meaning of law depends on exact wording, cross-references, and which of several similar-sounding instruments actually governs the situation at hand. EU climate and energy law makes this unusually difficult:

the Emissions Trading System directive of 2003 has been amended repeatedly, the Carbon Border Adjustment Mechanism borrows its allowance definitions from that directive, and dozens of implementing and delegated acts reuse the same vocabulary (*free allocation, carbon leakage, monitoring and reporting*) for legally distinct obligations. A retrieval system that returns a plausible but wrong instrument does not produce an approximate answer; it produces the wrong law. The stakes are highest in retrieval-augmented generation (RAG) pipelines, where the first-stage retriever determines whether citation-grounded generation is possible at all [13]. This paper evaluates that first stage.

The standard tension in legal retrieval is between lexical and semantic matching. BM25 performs well on formulaic legal text, where exact term overlap is a strong relevance signal [1], but a compliance officer asking whether her shipping company “has to pay for the pollution its vessels cause” shares almost no vocabulary with a provision on “surrender requirements in respect of maritime transport activities” (Directive 2023/959). Dense retrievers [19] bridge that vocabulary gap, but bring failure modes of their own: semantically similar passages from the wrong instrument can rank highly because legal drafting recycles structural boilerplate [11], and off-the-shelf embedding models often degrade on specialised corpora outside their training distribution [13], [18]. Hybrid sparse–dense fusion is the commonly recommended remedy [2], [3], [9]. Whether that recommendation holds for statute-level retrieval over a small, terminologically dense corpus is, however, an empirical question, and answering it fairly turned out to be harder than building the systems.

Our first evaluation attempt taught us why. In a pilot benchmark whose queries were written while consulting the corpus, every configuration looked excellent and the hybrid appeared to win outright; the strongest system reached a perfect hit rate. The queries had simply inherited the statutes’ vocabulary, so the benchmark was rewarding term overlap rather than retrieval quality, flattering BM25 in particular. We therefore rebuilt the benchmark under an explicit leakage-control protocol: every query was rewritten in the plain language of a non-expert, and the test set was extended with queries derived from written parliamentary questions that Members of the European Parliament submitted to the

Commission: real information needs, documented by their parliamentary reference numbers, whose official answers often name the very instruments a correct retrieval should return. To our knowledge, no prior work on hybrid legal retrieval has evaluated statute-level EU climate law, and none has controlled for query–corpus vocabulary leakage in this way.

We ask two questions. **RQ1.** Does hybrid sparse–dense retrieval improve effectiveness and reliability over BM25-only and dense-only baselines for EU Climate Law statute retrieval? **RQ2.** What failure modes remain for the strongest configuration, and do they indicate that first-stage retrieval is the bottleneck for grounded legal RAG in this domain?

Our contributions are: (i) a 102-query benchmark for statute-level EU Climate Law retrieval with binary CELEX-level relevance judgements and a sealed 53-query test set, constructed under the leakage-control protocol above, with 31 test queries traceable to documented European Parliament written questions; (ii) a reproducible corpus of 1,156 article-level provisions from 72 EU climate instruments, extracted through a documented two-phase SPARQL pipeline over the CELLAR repository; and (iii) a three-way comparison of BM25, a pure dense bi-encoder, and weighted reciprocal rank fusion, with bootstrap confidence intervals, significance testing, and per-query failure analysis.

To preview the results: on the leakage-controlled test set, both the dense retriever and the hybrid significantly outperform BM25, but the hybrid is statistically indistinguishable from the dense retriever alone, and the direction of that comparison reversed as the test set grew, a caution against reading small-sample differences between strong systems. The remaining failures are ranking errors on provisions buried inside large amending instruments, not gaps in corpus coverage, which points to corpus preprocessing rather than retrieval strategy as the next lever for improvement.

The remainder of this paper is structured as follows. Section II reviews related work on lexical, dense, and hybrid legal retrieval. Section III describes the methodology: corpus construction, the two retrievers, and the fusion method. Section IV details the benchmark construction protocol, metrics, and experimental configuration. Section V presents the performance evaluation, including significance testing and per-query failure analysis; Section VI discusses the implications; and Section VII concludes with limitations and future work.

II. RELATED WORK

A. Lexical vs. Dense Retrieval in Legal IR

Mori et al. [1] demonstrate that BM25 remains a competitive baseline for formulaic legal texts, where exact term overlap is a reliable relevance signal and dense models underperform because semantic similarity glosses over legally binding phrasing. Zheng et al. [7] find the opposite failure direction: lexical models degrade on reasoning-intensive queries, which semantic models handle more gracefully. The LRAGE evaluation framework [13] confirms that off-the-shelf dense retrievers underperform BM25 on legal retrieval tasks without domain-specific adaptation. Li et al. [21] present a contrasting

case: on large-scale Chinese legal case retrieval (LeCaRDv2), BM25 consistently outperforms off-the-shelf dense models, indicating that the balance between the paradigms is corpus- and language-dependent; our results suggest it also depends on how the evaluation queries are phrased.

B. Hybrid Retrieval

Rayo et al. [3] show that combining BM25 with fine-tuned sentence transformers substantially improves Recall@10 and MAP@10 on regulatory corpora. LegalRAG [9] demonstrates that hybrid systems outperform single-mode systems for multilingual legal texts. HyPA-RAG [2] introduces adaptive parameter adjustment for hybrid retrieval in AI legal and policy applications. SPLADE [4] takes a different route, learning sparse representations that close the vocabulary gap inside a single inverted index; we stay with classical BM25 because it needs no training data and keeps the lexical signal fully interpretable. On the fusion method itself, Cormack et al. [14] established that rank-based fusion outperforms score-based combination without requiring score normalisation, while Bruch et al. [20] show analytically that RRF is sensitive to its parameters and that settings do not transfer reliably across collections, a finding our validation sweep reproduces on a much smaller statutory corpus.

C. Retrieval Reliability and Legal RAG

Reuter et al. [11] identify Document-Level Retrieval Mismatch (DRM) as a systematic failure mode of dense retrievers on large legal datasets, where shared structural boilerplate causes globally irrelevant documents to rank highly. Ho et al. [12] show that doctrinal accuracy requires structural legal signals beyond semantic similarity. CBR-RAG [5] and LexRAG [6] demonstrate that multi-turn legal grounding requires retrieval robust to topic drift and implicit definitional dependencies. Multi-stage architectures combining embedding models with LLM re-ranking have proven effective in the COLIEE legal retrieval competition [10], and LexDrafter [8] applies retrieval-augmented generation directly to EU legislative documents for terminology drafting; this is the closest prior work to our corpus domain, though it does not evaluate first-stage retrieval quality.

III. METHODOLOGY

Fig. 1 shows the end-to-end pipeline: corpus construction from CELLAR, dual indexing, and query-time fusion.

A. Corpus Construction

We extract the corpus from the EU CELLAR repository in two phases. A SPARQL query first retrieves every work URI associated with any of 21 EuroVoc climate and energy concept identifiers, which yields the 72 instruments; per-document manifestation queries then resolve the Formex XML (`fm:x4`) download URL for each instrument, falling back to the HTML manifestation for the few documents that lack one. Formex is the Publications Office’s structured drafting format, so parsing it (with `lxml`) recovers the legal structure of each act directly instead of reverse-engineering it from rendered HTML.

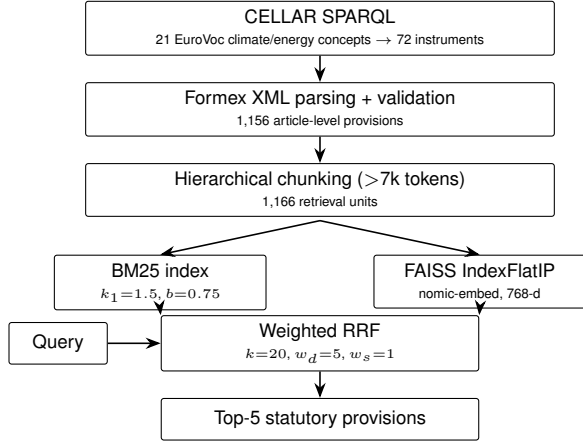


Fig. 1: System pipeline: two-phase CELLAR extraction, dual sparse/dense indexing over 1,166 retrieval units, and query-time weighted reciprocal rank fusion. Both retrievers return their top-100 candidates concurrently.

The retrieval unit is the legislative article. This is a deliberate choice rather than a convenience: EU articles are drafted as self-contained units (each states one obligation, definition, or procedure), so they align with what a legal user actually needs retrieved, and they spare us the arbitrary window-size decisions of token-based chunking. Concretely, each `<ARTICLE>` element becomes one unit, with text reconstructed from `<ALINEA>` leaf nodes to avoid duplicated nesting; every record is validated against a schema, and placeholder articles are discarded. The exception is amending instruments, whose Article 1 can run to tens of thousands of tokens because it contains every amendment the act makes. Five such articles exceed a 7,000-token limit (chosen to leave headroom below the dense encoder’s 8,192-token window) and are split at the natural amendment-point boundaries; the resulting sub-chunks inherit the parent `celex_id`, so retrieval results always resolve to a citable instrument.

The final corpus contains **1,156** article-level provisions (1,166 retrieval units after splitting) from **72** CELEX instruments covering ETS, CBAM, Taxonomy, LULUCF, F-Gas, Maritime MRV, ESR, the European Climate Law, Energy Efficiency, CSDDD, and the Governance Regulation.

B. Sparse Retrieval - BM25

The lexical retriever is BM25Okapi [15] (`rank-bm25` 0.2.2) with the standard parameters $k_1=1.5$ and $b=0.75$. The only customisation is the tokeniser, which is built for statutory text: hyphenated terms of art such as *net-zero* survive as single tokens, and we apply neither stemming nor stopword removal. Both omissions are intentional. EU definitions are terminologically precise: *allowance* and *allowances* can appear in differently scoped provisions. Stopword removal, in turn, would break fixed legal phrases such as *free of charge* and *not later than*, whose function words carry the legal meaning. At query time BM25 scores all 1,166 units and returns the top 100.

C. Dense Retrieval - nomic-embed-text-v1.5

The semantic retriever is `nomic-embed-text-v1.5` [16], a 768-dimensional bi-encoder with an 8,192-token context window, loaded via `sentence-transformers`. Three properties drove the selection. It is fully open-weight, which keeps the pipeline reproducible without API access. Its long context window embeds EU articles whole (only the five oversized amendment articles described above need splitting), where a conventional 512-token encoder would silently truncate roughly a fifth of this corpus. Most importantly for the experimental design, it is a *pure* bi-encoder: models trained jointly for dense, sparse, and late-interaction retrieval (such as BGE-M3) already encode lexical signals in their dense vectors, which would muddy any conclusion about what BM25 adds on top.

Following the model’s asymmetric training scheme, queries are prefixed with `"search_query:"` and articles with `"search_document:"` before encoding. Corpus vectors are L2-normalised and stored in a FAISS [17] `IndexFlatIP`, where inner product on normalised vectors equals cosine similarity; at 1,166 vectors, exact search is trivially fast and no approximate index is needed. At query time, FAISS returns the top 100 nearest neighbours.

D. Reciprocal Rank Fusion

Both retrievers run concurrently on each query (`ThreadPoolExecutor`, two workers), and their ranked lists are merged by weighted Reciprocal Rank Fusion [14]. Each candidate document d receives the score

$$\text{RRF}(d) = \frac{w_s}{k + \text{rank}_s(d)} + \frac{w_d}{k + \text{rank}_d(d)} \quad (1)$$

where rank_s and rank_d are the document’s positions in the sparse and dense lists (∞ if absent), k dampens the influence of any single top-ranked result, and w_s , w_d weight the two retrievers. The intuition of (1) is that a document ranked well by both retrievers should beat a document ranked first by only one; the weights let a systematically stronger retriever dominate ties. The top 5 fused results form the system’s output (and, in a full RAG deployment, the context passed to generation).

We fuse ranks rather than raw scores because the two score scales are incompatible: BM25 produces unbounded positive reals while cosine similarity lies in $[-1, 1]$, and any direct combination smuggles in scale assumptions that rank-based fusion avoids entirely [14]. The operating point $k=20$, $w_s=1$, $w_d=5$ was chosen by grid search on the 49-query validation split; the sealed 53-query test set was evaluated only after these values were frozen. Fig. 2 shows the validation landscape: ranking quality improves as the dense retriever is up-weighted, degrades sharply at large k , and the deployed configuration is the highest-MRR cell among those reaching $\text{HR}@5=1.000$ on validation.

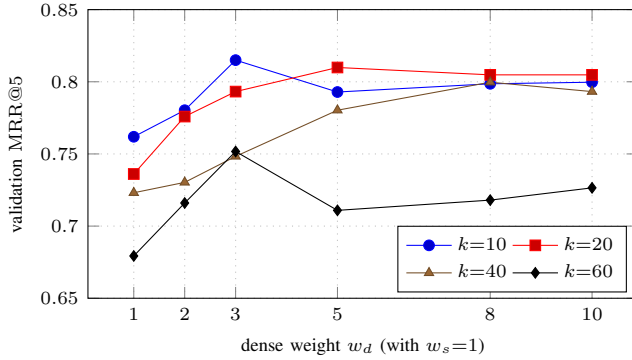


Fig. 2: Validation-split MRR@5 across the RRF grid. The deployed configuration ($k=20$, $w_d=5$; MRR@5=0.810) is the best-ranking cell that also achieves HR@5=1.000. The literature-default region ($k=60$, low w_d , bottom-left) performs worst.

IV. EXPERIMENTAL SETUP

A. Benchmark

The benchmark contains 102 queries covering all eleven EU Climate Law instrument families, constructed in two stages. Unlike broad multi-task legal benchmarks that aggregate hundreds of tasks across jurisdictions [22], it targets a single tightly bounded corpus, which is what makes statute-level retrieval evaluation and leakage control tractable. An initial set of 71 queries was written in deliberately natural, non-statutory language: an early pilot in which queries paraphrased corpus wording produced near-perfect scores for all systems (HR@5=1.0), indicating query-corpus vocabulary leakage rather than retrieval quality; all queries were therefore rewritten to express the underlying information need without reusing statutory vocabulary. These 71 queries were split 70/30 by stratified difficulty (seed 42) into a 49-query validation set (used exclusively for fusion-parameter tuning) and a 22-query test set. The test set was subsequently expanded with 31 queries derived from written questions submitted by Members of the European Parliament to the Commission: each parliamentary question was screened for answerability from legislation, paraphrased into plain language, and mapped to its relevant CELEX instruments, with the parliamentary reference number (e.g., E-002271/2024) recorded per query as an audit trail. In many cases the Commission’s official answer cites the relevant instruments, providing independent corroboration of the relevance judgement. Corpus membership was verified for every mapping before a query was accepted, so no query requires an instrument absent from the corpus. All reported results use the resulting sealed **53-query test set**; the validation set was never evaluated against reported metrics. To give a flavour of the derivation: parliamentary question E-004153/2025 asks whether a ship calling at a mainland port en route to an outermost region owes allowances for the first leg of the voyage; this becomes the plain-language query “*A ship sails from the mainland, stops at another mainland port, then continues to one of the EU’s remote overseas regions — is the whole journey spared the carbon charge, or only the*

TABLE I: Benchmark composition. “ ≥ 2 rel.” counts queries with two or more relevant instruments; “instr.” counts distinct instruments used as relevance targets.

Subset	n	standard	hard	≥ 2 rel.	instr.
Validation	49	36	13	29	44
Test (original split)	22	16	6	9	26
Test (EP-derived)	31	19	12	13	22
Test (total)	53	35	18	22	37
All queries	102	71	31	51	62

last leg?”, mapped to Directive 2023/959, whose derogation the Commission’s answer cites.

Each query carries a difficulty tag used for stratification: *hard* queries share essentially no vocabulary with the statutory text (e.g., “the don’t-make-anything-else-worse test” for the Taxonomy’s *do no significant harm* principle), while *standard* queries are naturally phrased but moderately terminological. Table I summarises the composition. Relevance judgements target 62 of the 72 corpus instruments; the ten never used as targets are annex-replacement amendment acts with no self-contained content, excluded as relevance targets by construction.

Relevance is binary at the CELEX instrument level: a document is relevant if it contains the provision that directly and materially answers the query. CELEX identifiers with corrigendum suffixes (e.g., 32003L0087R(02)) are normalised by stripping the R(. . .) suffix before matching.

B. Metrics

We report three metrics at cut-off 5, matching the number of provisions a downstream RAG stage would receive as context. HR@5 measures coverage: the fraction of queries for which at least one relevant instrument appears in the top 5 at all. For grounded generation this is the make-or-break number: a miss here cannot be repaired downstream. MRR@5 measures how high the first relevant result lands, which matters because generation quality degrades when the correct provision sits below distractors. NDCG@5 credits finding *all* relevant instruments, not just the first: 22 of the 53 test queries have two or more relevant instruments (Table I), and MRR is blind to everything after the first hit. Gains are binary, and because retrieval is article-level, several articles of the same instrument can occupy the top 5; each relevant instrument is therefore credited once, at its highest-ranked occurrence, since crediting duplicates would inflate NDCG above 1.

With only 53 test queries, point estimates alone would be easy to over-read, so we quantify uncertainty twice. 95% confidence intervals are estimated by percentile bootstrap over queries (5,000 resamples, seed 42), which assumes nothing about the score distribution. Pairwise comparisons between systems use the Wilcoxon signed-rank test on per-query MRR ($\alpha=0.05$, two-tailed, $n=53$), a paired, non-parametric test

appropriate for skewed per-query scores, and we report Cohen’s d alongside p -values so that effect sizes are visible even where significance is not reached.

C. Hardware and Software

All reported results were produced on a consumer laptop (AMD Ryzen 7 5700U, CPU-only, Windows 11): deliberately modest hardware, since one claim of this work is that the full pipeline is reproducible without GPUs or paid APIs. The software stack is Python 3.14.2 with `sentence-transformers` 5.5.1, `faiss-cpu` 1.14.2, `rank-bm25` 0.2.2, and `pydantic` 2.13.4; all versions are pinned in the repository, and the combined indices occupy under 10 MB.

V. PERFORMANCE EVALUATION

A. Overall Effectiveness

Table II reports the headline comparison on the sealed 53-query test set.

The clearest result is the gap to BM25. Both semantic configurations outperform it by more than twenty points of HR@5, and both differences are statistically significant: hybrid vs. BM25 at $p=0.0005$ (Cohen’s $d=0.579$, medium effect) and dense vs. BM25 at $p=0.004$ ($d=0.433$, small effect). On a leakage-controlled benchmark, lexical retrieval alone misses roughly three queries in ten.

Between the two semantic configurations the picture is a statistical tie. Dense retrieval posts the best HR@5 (0.925 vs. 0.906) while the hybrid posts the best MRR@5 (0.773 vs. 0.743) and NDCG@5 (0.721 vs. 0.709), but the Wilcoxon test finds no significant difference ($p=0.614$, $d=0.120$, negligible), and the bootstrap confidence intervals of the two systems overlap substantially on every metric (e.g., MRR@5: hybrid [0.666, 0.868] vs. dense [0.639, 0.836]; NDCG@5: hybrid [0.627, 0.842] vs. dense [0.621, 0.791]). The honest answer to RQ1 is therefore split: fusion delivers a significant improvement over the lexical baseline, but no measurable improvement over a strong dense retriever used alone.

B. Effect of Query Difficulty

Table III splits the test set by the difficulty tag introduced in Section IV: *standard* queries are naturally phrased but moderately terminological, while *hard* queries share essentially no vocabulary with the statutes.

The split localises exactly where lexical retrieval breaks. On standard queries BM25 is usable (MRR@5=0.611); on hard queries its MRR@5 nearly halves to 0.366, while dense retrieval degrades far more gently (0.784 to 0.661). The gap between the paradigms therefore *widens* with vocabulary mismatch, from 0.17 to 0.30 in MRR@5, which is the designed purpose of the hard tier and direct evidence for the mechanism claimed throughout: BM25’s failure mode is vocabulary, not ranking. The same split also explains where fusion helps at all: the hybrid’s MRR@5 advantage over dense is concentrated on standard queries (+0.037) and shrinks on hard ones (+0.019), consistent with BM25’s signal being trustworthy only when the query shares at least some statutory vocabulary.

C. Quantifying Vocabulary Leakage

The pilot benchmark’s original corpus-derived queries were preserved, which permits a controlled experiment: the same 71 information needs, with identical relevance judgements, evaluated under both their original statutory wording and their leakage-controlled rewriting. Nothing else varies: corpus, indices, and systems are held fixed. Table IV reports the result.

Under leaky wording, all three systems sit within three points of one another and the benchmark cannot distinguish them. Rewriting the queries costs BM25 0.254 of HR@5 and 0.290 of MRR@5, while the dense retriever loses only 0.028 and 0.085 respectively; leakage inflates lexical retrieval roughly three and a half times more than semantic retrieval, manufacturing a parity that does not survive realistic phrasing. Two caveats: these 71 pairs include the 49 validation queries, so the absolute values are not comparable with Table II and carry no held-out guarantee; and no tuning was performed on either wording of this set beyond the validation use already described. The comparison is between wordings, not systems, and for that purpose the pairing is exact.

D. A Worked Example

Table V shows the actual top-5 lists for one hard test query: “*Is there any leeway in the EU’s land-carbon rules for a country whose forests get wrecked by wildfires or bark beetles?*” The relevant instruments are the LULUCF Regulation (2018/841) and its 2023 amendment (2023/839), whose flexibility provisions speak of “natural disturbances”; none of the query’s content words appears in the statute. With nothing lexical to anchor on, BM25 is confidently wrong across two unrelated regimes, filling its list with the European Green Bond Standard and CBAM instruments on the strength of incidental term matches. The dense retriever maps the paraphrase onto the natural-disturbance provisions directly, placing both relevant instruments in its top five; the dense-weighted fusion preserves them.

E. Hard Failure Analysis

RQ2 asks what failures remain; Table VI breaks down per-query outcomes across the three systems.

Dense retrieval recovers 13 queries that BM25 misses entirely, confirming that semantic generalisation is the primary source of improvement over lexical retrieval. Fusion is nearly lossless with respect to the dense ranking: the hybrid preserves 48 of dense retrieval’s 49 hits, losing exactly one query where BM25 interference displaces the relevant document from the top 5. Conversely, BM25 contributes exactly one unique hit that dense retrieval misses, and the dense-favouring fusion weight ($w_d=5$) suppresses the BM25 signal too strongly for the hybrid to preserve it. Three queries are missed by all three systems, and the hybrid fails five in total. Manual inspection shows that every failed query’s relevant instrument *is* present in the corpus: these are ranking failures, not coverage gaps. Four distinct causes account for the five hybrid failures: two ETS2 queries whose relevant provisions sit inside the sub-chunks of Article 1 of Directive 2023/959, where amendment

TABLE II: Retrieval effectiveness on the 53-query test set. Bootstrap 95 % CIs in parentheses for HR@5. $\dagger p=0.0005$ vs. BM25; $\ddagger p=0.004$ vs. BM25.

System	HR@5	MRR@5	NDCG@5
BM25	0.698 (0.566–0.811)	0.528	0.494
<i>nomic-embed-text-v1.5</i>	0.925 (0.849–0.981) $\ddagger$	0.743	0.709
Hybrid RRF	0.906 (0.811–0.981) $\dagger$	0.773	0.721

TABLE III: Results by query difficulty (standard $n=35$, hard $n=18$).

System	Subset	HR@5	MRR@5	NDCG@5
BM25	standard	0.743	0.611	0.559
	hard	0.611	0.366	0.367
<i>nomic-embed-text-v1.5</i>	standard	0.943	0.784	0.738
	hard	0.889	0.661	0.654
Hybrid RRF	standard	0.943	0.821	0.754
	hard	0.833	0.680	0.655

TABLE IV: Paired leakage experiment: 71 identical information needs and relevance judgements under corpus-derived (leaky) versus plain-language (controlled) wording.

System	Leaky wording		Controlled wording	
	HR@5	MRR@5	HR@5	MRR@5
BM25	0.958	0.846	0.704	0.556
<i>nomic-embed-text-v1.5</i>	0.986	0.870	0.958	0.785
Hybrid RRF	0.986	0.876	0.972	0.795

TABLE V: Top-5 retrieved instruments for the wildfire/bark-beetle query. Relevant instruments in **bold**; repeated entries are different articles of the same instrument.

Rank	BM25	<i>nomic-embed-text-v1.5</i>	Hybrid RRF
1	EuGB 2023/2631	LULUCF 2023/839	LULUCF 2023/839
2	CBAM am. 2025/2083	LULUCF 2023/839	LULUCF 2023/839
3	CBAM 2023/956	LULUCF 2018/841	Nat. Rest. 2024/1991
4	CBAM 2023/956	Nat. Rest. 2024/1991	LULUCF 2018/841
5	EuGB 2023/2631	LULUCF 2018/841	LULUCF 2018/841

TABLE VI: Per-query hit pattern breakdown ($n=53$).

Category	n	Fraction
All three systems hit	36	0.679
Dense and hybrid hit; BM25 misses	12	0.226
Only BM25 hits	1	0.019
Only dense hits (hybrid misses)	1	0.019
All three miss	3	0.057

boilerplate dominates the embedding; one high-abstraction conceptual query about the legal character of the Taxonomy framework; one query whose colloquial phrasing (“*suppliers wrecking the environment*”) sits far from the statutory register of due-diligence obligations; and one confusion between adjacent EU funding instruments. Answering RQ2: the residual failures are concentrated in corpus structure (amendment-act

sub-chunks) and extreme vocabulary mismatch, not in the retrieval strategy itself.

F. Ranking Quality

Where does the hybrid’s aggregate MRR@5 advantage (+0.031) come from? A per-query comparison on the 48 queries that both semantic systems hit shows it is not a systematic pattern: the two systems place the first relevant result at the same rank on 31 queries, the hybrid ranks it higher on 8, and dense ranks it higher on 9. The aggregate edge instead comes from magnitude: the hybrid’s improvements are more often full promotions to rank 1 (38 queries place a relevant instrument at rank 1, versus 34 for dense and 23 for BM25), typically where BM25’s exact-term signal confirms the correct instrument that *nomic-embed-text-v1.5* had placed just below a semantically similar alternative. This is the mechanism the complementarity hypothesis predicts, but

at this corpus size it operates on a handful of queries, enough to move the mean but not enough to reach significance.

VI. DISCUSSION

Why does fusion not pay off here? For fusion to improve on its stronger component, the weaker component must rescue queries the stronger one misses. On this benchmark that almost never happens: dense retrieval misses only four queries, and BM25 recovers exactly one of them. With so little unique lexical signal to add, fusion can at best re-rank what dense already found (precisely what the per-query analysis shows), while the dense-favouring weight needed to stop BM25’s errors from corrupting the ranking also suppresses its one genuine save. This is not a general argument against hybrid retrieval. The reported hybrid gains in the literature typically arise where the dense component is weaker relative to the corpus (older or smaller encoders, broader domains) or where the corpus is large enough that the retrievers diverge more often; on a small, terminologically dense statutory corpus, a strong modern bi-encoder simply leaves BM25 too little room to help.

Small samples mislead. The direction of the hybrid-vs-dense comparison reversed as the test set grew (dense led on MRR at $n=22$; the hybrid leads at $n=53$) while remaining non-significant at both sizes. Had we reported the $n=22$ numbers, the natural narrative would have been “fusion hurts”; at $n=53$ it would read “fusion helps.” Both readings are noise. Legal IR benchmarks in this size range are common, and comparisons drawn from them without significance testing should be treated with corresponding suspicion. That includes comparisons in our own earlier drafts.

Evaluation design decides the apparent winner. The sharpest lesson of this work came before any system comparison, and the paired experiment of Table IV quantifies it: holding corpus, indices, systems, and relevance judgements fixed, merely rewording the queries moves BM25 by -0.29 MRR@5 but the dense retriever by only -0.09 . Under corpus-derived wording the benchmark reports a three-way tie; under realistic wording it reports a twenty-five-point spread. Nothing about the retrievers changed; only the words did. Since annotators who write evaluation queries while reading the corpus will naturally reuse its vocabulary, we suspect this inflation is present in other specialised-domain benchmarks, and that it systematically flatters lexical and hybrid systems. Externally sourced information needs, such as the parliamentary questions used here, are a cheap and auditable defence.

Practical implications. For a practitioner building statutory retrieval today, these results support a simple recipe: start with a strong open-weight bi-encoder; treat BM25 fusion as low-cost insurance rather than a required component (the combined indices occupy under 10 MB and the retrievers run concurrently, so the overhead is negligible, but so, on this evidence, is the benefit); and if fusion is used, tune it. The literature default of $k=60$ with equal weights [14] degrades validation performance from HR@5=1.000, MRR@5=0.810

(tuned: $k=20$, $w_d=5$) to HR@5=0.898, MRR@5=0.679, because with only $\sim 1,200$ candidate documents, BM25’s lexical errors sit close enough to the top to displace dense rank-1 results. Parameters calibrated on TREC-scale collections do not transfer to small specialised corpora, consistent with the parameter-sensitivity analysis of Bruch et al. [20]. Finally, the failure analysis locates the remaining headroom not in retrieval strategy but in corpus preprocessing: indexing consolidated instrument versions would dissolve the amendment-act sub-chunks responsible for the largest cluster of residual failures.

VII. CONCLUSION

This paper set out to test whether hybrid sparse–dense fusion improves statute-level retrieval for EU Climate Law, and found that answering the question fairly required building the evaluation as carefully as the system. We contributed a reproducible 1,156-provision corpus of 72 EU climate instruments, a 102-query benchmark constructed under an explicit leakage-control protocol, 31 of whose test queries derive from documented European Parliament written questions, and a three-way comparison of BM25, a pure dense bi-encoder, and weighted reciprocal rank fusion with significance testing throughout.

The findings are easy to state. Semantic retrieval is decisively better than lexical retrieval on realistically phrased legal queries: both dense-only and hybrid beat BM25 by wide, statistically significant margins. Between the two, there is no measurable difference; fusion acts as a cheap safety net rather than a necessary ingredient. And the pilot-to-controlled swing showed that benchmark construction, not retrieval architecture, was the largest single factor in which system appeared to win, which we regard as the paper’s most transferable lesson. The remaining failures cluster on provisions buried in amending instruments, so corpus preprocessing, not a new retriever, is the most promising next step for practitioners.

Several limitations bound these conclusions. The test set, at 53 queries, yields wide confidence intervals, and a null result at this size cannot exclude a small true effect. Relevance is judged at the instrument (CELEX) level rather than the article level, which is lenient toward systems that find the right act but the wrong provision. Query paraphrases and relevance mappings were produced with generative-AI assistance and verified by the authors; independent second-annotator validation with inter-annotator agreement is in progress but not reflected in the reported numbers. Finally, the study covers one legal domain, one language, and one dense encoder, so the “dense is enough” conclusion should be read as a claim about strong modern bi-encoders on small statutory corpora, not about dense retrieval in general.

Future work follows directly from these limitations: completing the independent annotation and reporting Cohen’s κ ; extending the relevance judgements to article level; indexing consolidated instrument versions to dissolve the amendment-article failure cluster; scaling the test set through the European Parliament’s open data interface, from which our query-derivation protocol can be largely automated; and evaluating

the generation stage of the RAG pipeline, for which the Commission’s official answers to the source parliamentary questions provide ready-made, citable reference answers.

GENERATIVE AI DISCLOSURE

Generative AI tools (Claude, Anthropic) were used solely for proofreading and improving clarity of writing. All research ideas, technical content, and academic contributions are entirely our own.

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